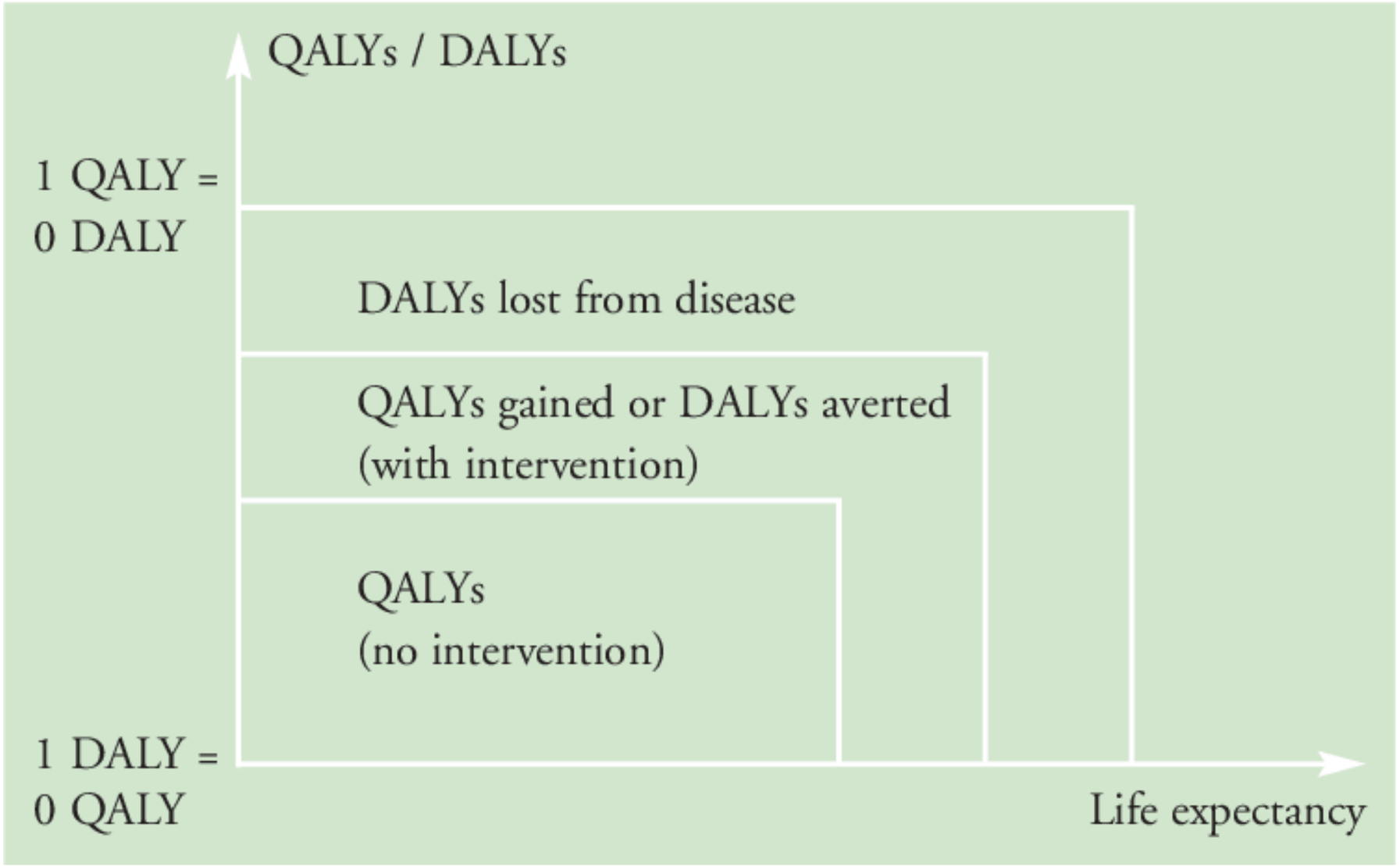
Lesson 6: Experienced burden of disease

Case study of a quantitative research project

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Introduktion til Sundhedsdata · Danish Centre for Health Economics (DaCHE), University of Southern Denmark

2026-10-06

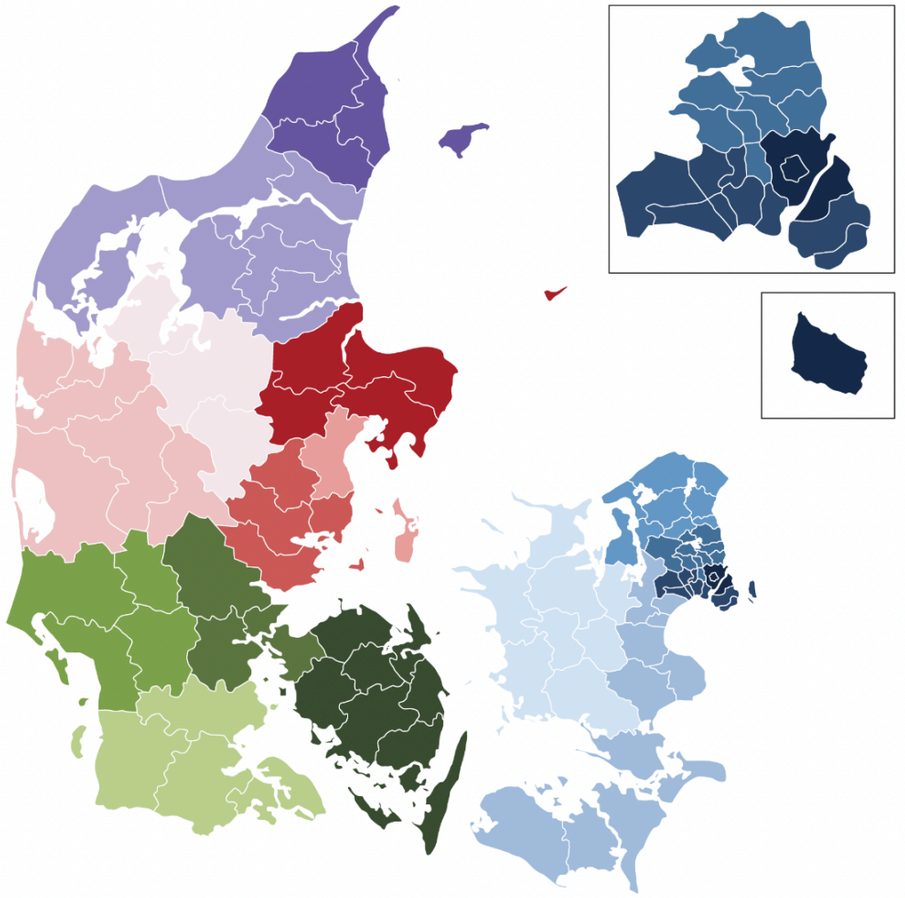
Priority setting in health



Fox-Rushby, Julia. "Disability-adjusted life years (DALYs) for decision-making?: an overview of the literature." (2002).

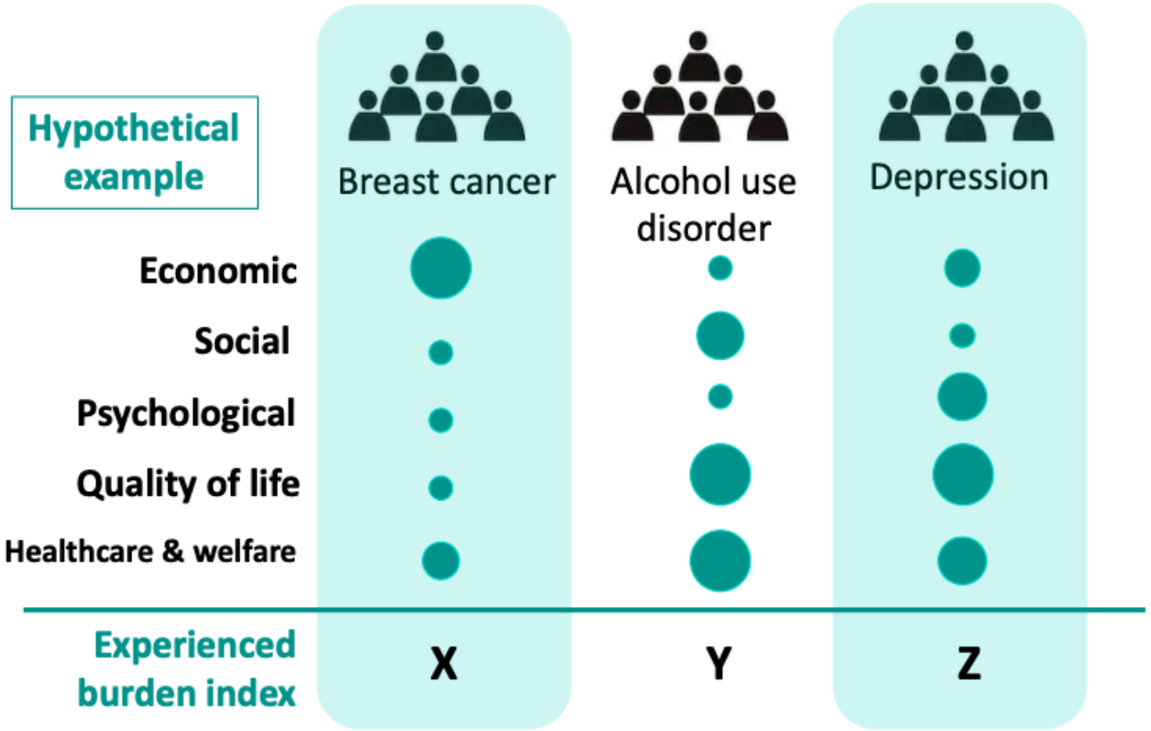
Priority setting in Denmark

- Historically: region-level prioritization based on general principles
- Danish Medicines Council: principled recommendations of coverage for pharmaceutical treatment
  - 2021: incorporation QALYs into assessment
- 2024: National Priority Council
  - Broader scope than DMC
  - Address demographic transition, labor constraints, regional and municipal variation



Gap: non-health consequences of disease

- Economic, social, psychological, consequences of disease not captured in DALY/QALY
- Hypotheses:
  - These impacts are substantial, even in Denmark
  - These may not correlate to morbidity/mortality patterns by disease



Experienced burden of disease in Denmark

- We aim to estimate longitudinal economic and social impacts of hospital-diagnosed mental and physical disease across the Danish population from an individual perspective.
  - Comprehensive: population-level, inclusive
  - Long-term: 10 year follow-up, 27 years of data
  - Comparative: across and within diseases
  - Rigorous but not causal
    - Measurement bias for exposure
    - Confounding

Identification

Exposure

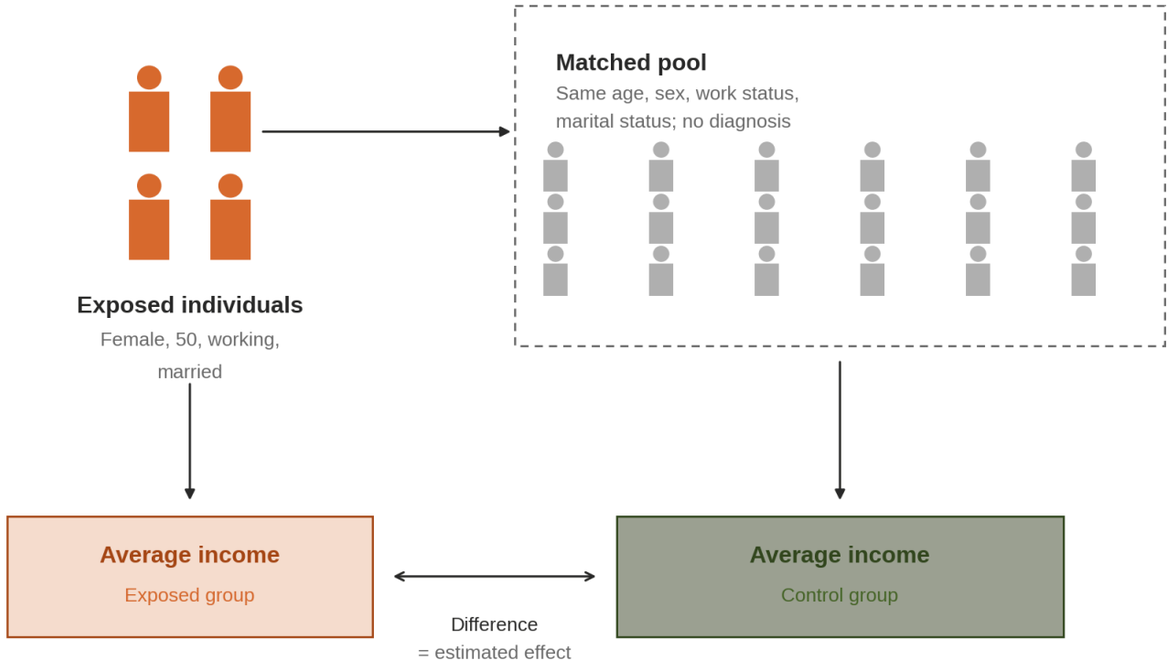
- 1st hospital diagnosis of disease (inpatient, outpatient, psychiatric)
- Mental dx:
  - Depression, alcohol use disorder (AUD), eating disorders, bipolar disorder
- Somatic dx:
  - Breast cancer, stroke, spinal disc disorders, epilepsy, type 1 diabetes, inflammatory bowel disease (IBD)

Outcomes

- Part 1: personal disposable income
- Part 2:
  - Biological/adoptive parenthood
  - Household composition (children living in home)
- Part 3:
  - Total net wealth, all sources
  - Total savings and liquid assets
  - Property ownership (binary)
  - Relationship formation & dissolution

Red: 1st paper, income loss Green: 2nd paper, parenthood Orange: planned future work

Study design & methods



Matching variables:

Demographics

- Age, sex
- Marital status
- Household size

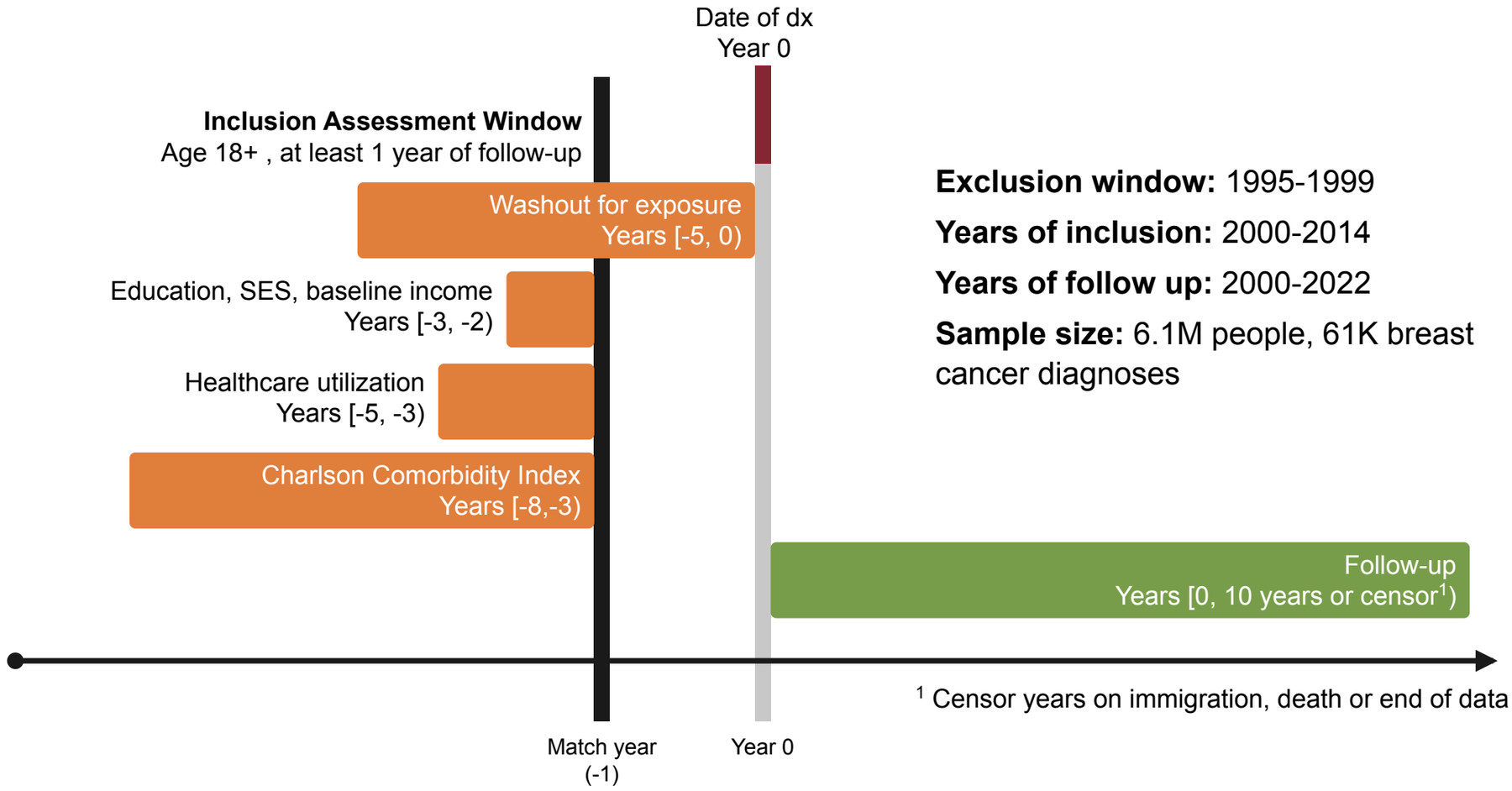
Socio-economics

- Baseline income quantile
- Educational attainment
- Labor market attachment

Baseline health

- Hospital contacts
- Charlson comorbidities

Study design & methods



Exclusion window: 1995-1999

Years of inclusion: 2000-2014

Years of follow up: 2000-2022

Sample size: 6.1M people, 61K breast cancer diagnoses

[1] Censor years on immigration, death or end of data

Estimating income loss

Conditional income losses:

$$\Delta_{g,i} = \overline{Y}^{C}_{g,i} - \overline{Y}^{E}_{g,i}$$

- Losses within each match subgroup and year

Total "real" income loss:

$$AILX_i = \sum_g p^{E}_{g,i} \cdot \Delta_{g,i}$$

- Equivalent to ATT

Total "standardized" income loss:

$$AIL_i = \sum_g p^{D}_{g,i} \cdot \Delta_{g,i}$$

- Equivalent to ATE, or age-sex standardization

Where:

- $\overline{Y}^{C}_{g,i}$ is the control income in group g at time i
- $\overline{Y}^{E}_{g,i}$ is the exposed income in group g at time i
- $p^{E}_{g,i}$ is the prop of exposed population in group g at time i
- $p^{D}_{g,i}$ is the prop of Danish population in group g at time i

Data management

Step	Code processes	Rerun reasons	# reruns
Raw data prep	Recode categories Merge raw data Renaming	Rare	8
Matching	Define match groups by variable sets `df %>% mutate(group_id = cur_group_id(), .by = c(var1, var2))`	Any time I change a match variable	28
Merging future outcomes	Define exposed vs. control Merge 10 years of follow-up by match group Match group level aggregation	Change in exposure Change in match variable	37
Aggregation	Aggregate along subgroup combinations	Any time I need different aggregations	~11

Data management

```
/code
  /01_data_prep
    01_hospital.R
    02_populations.R
    03_family_variables.R
    04_data_prep_complete.R
  /02_analysis
    01_matching.R
    02_identify_exposed.R
    03_run_estimation.R
    04_bootstrap.R
    match_functions.R
  /03_results
    .....
```

```
/data
  /01_prepped
    20250401
    20251001
    ...
  /02_matched
    20250122
    20250123
    ...
  ...
```

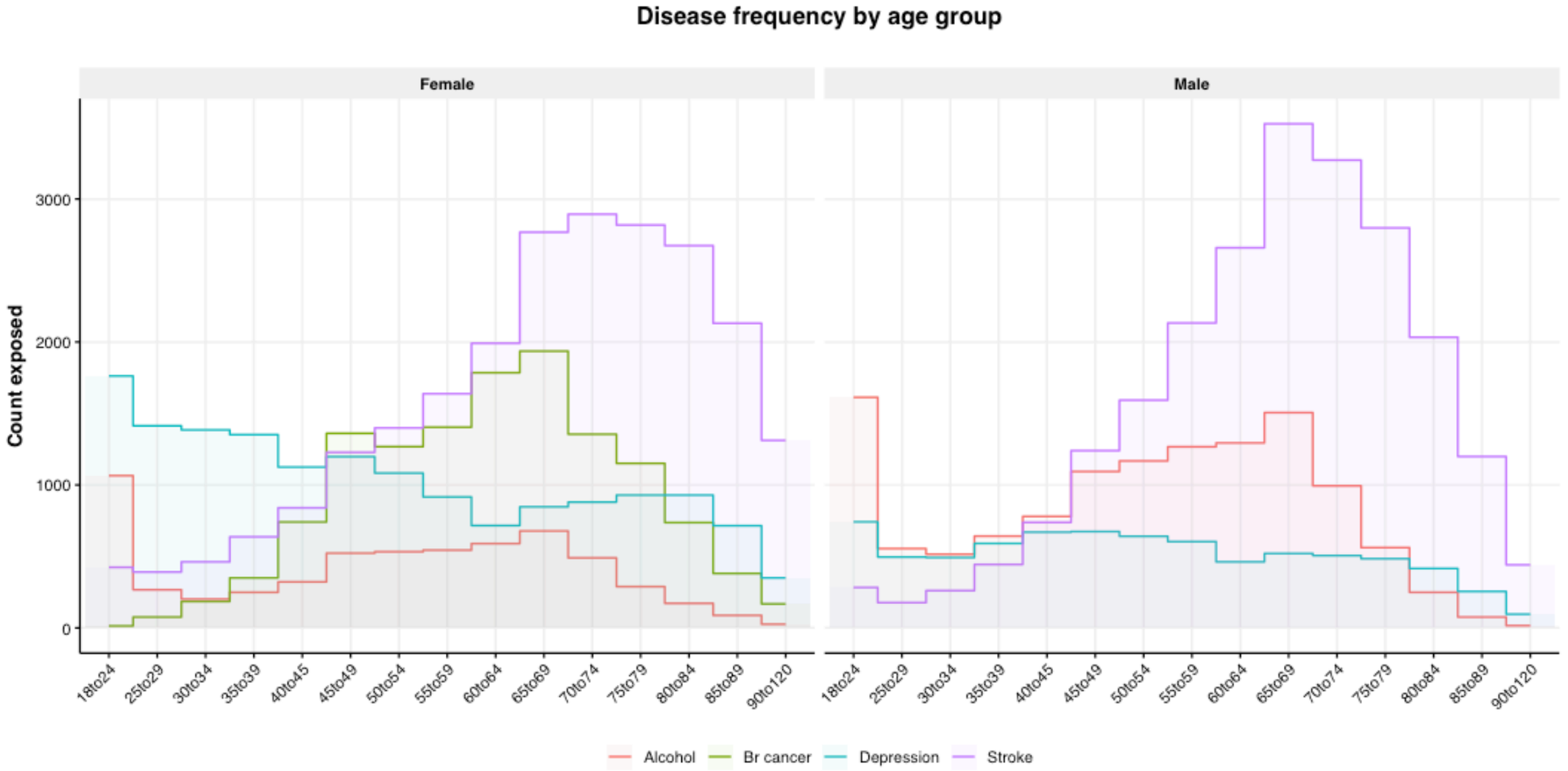
Data visualization

- Questions:
  - What do the different disease populations look like?
  - How do they differ?
  - What are the average outcomes of exposed, control groups?
  - What are the impacts of disease?
- Presentation:
  - Exploration vs. explanation

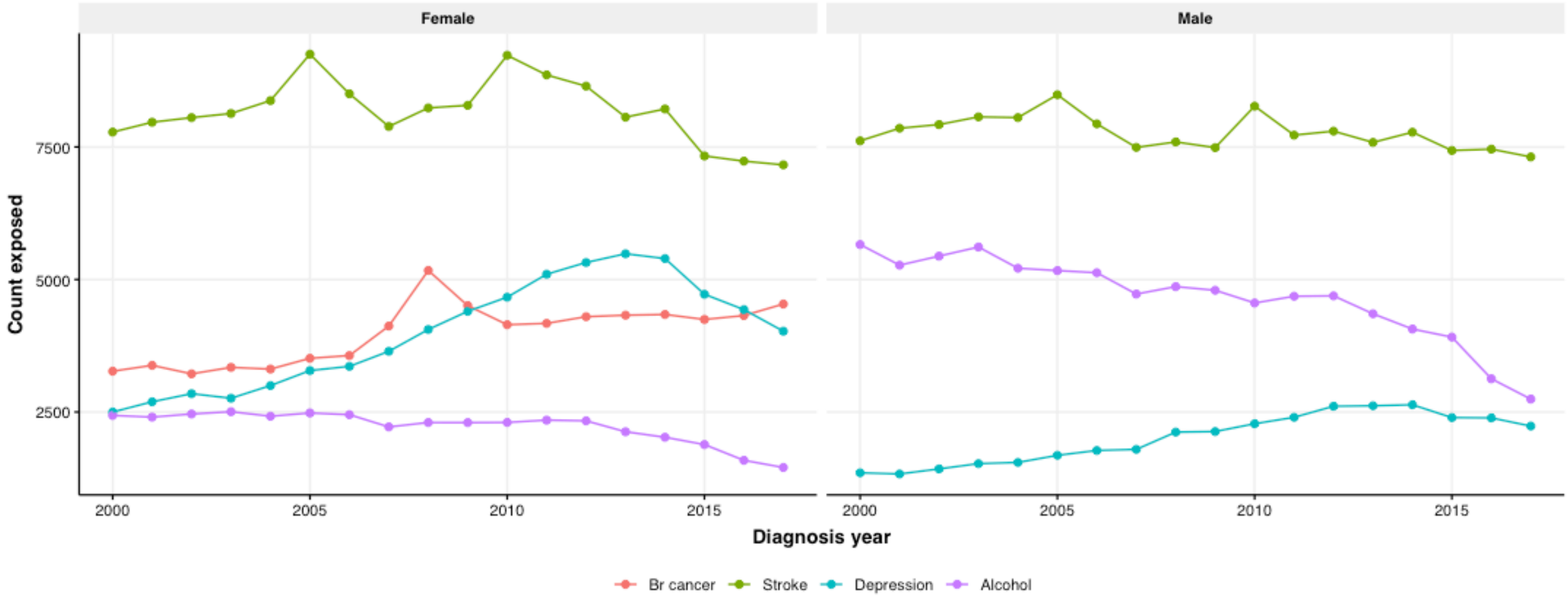
Data visualization

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Disease population comparison – age at exposure



Disease population comparison – diagnosis year



Disease population comparison – labor status

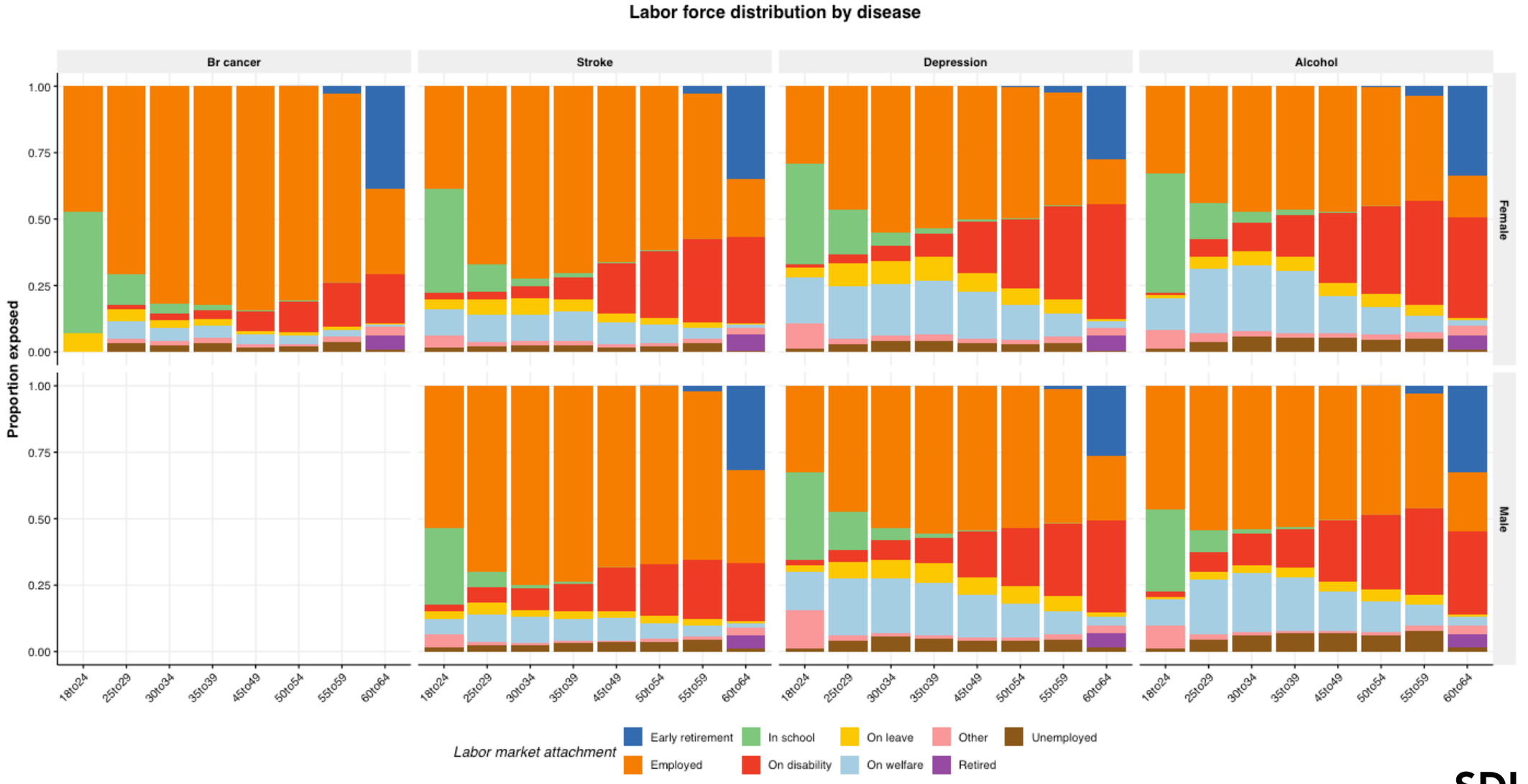


Table 1

- Standard population characteristics table in health econ, epi papers
- Describes general population, your sample, your comparison
- Often grouped
- Useful for exploration and explanation

Table 2. Characteristics of the Danish Population and Exposed Cohorts, 2012[a]

Characteristic	General population (N = 2 911 225), %	Diagnosis, % (SE)			
		Depression (n = 10 284)	Alcohol use disorder (n = 5847)	Stroke (n = 6539)	Breast cancer (n = 2230)
10-y Mortality rate, %	3.5	6.9	18.0	14.0	16.1
Sex					
Female	49.9	61.7 (0.24)	33.4 (−0.34)	46.8 (−0.06)	100 (1.42)
Male	50.1	38.3 (−0.24)	66.6 (0.34)	53.2 (0.06)	NA
Age group, y					
18-24	15.5	25.2 (0.24)	21.6 (0.16)	3.4 (−0.42)	0.8 (−0.56)
25-29	10.3	13.2 (0.09)	6.8 (−0.12)	4.0 (−0.25)	1.3 (−0.39)
30-34	9.6	9.3 (−0.01)	6.3 (−0.12)	4.1 (−0.22)	3.3 (−0.26)
35-39	10.8	10.3 (−0.02)	7.9 (−0.10)	5.5 (−0.19)	5.7 (−0.19)
40-44	11.6	10.1 (−0.05)	9.3 (−0.07)	9.6 (−0.06)	11.3 (−0.01)
45-49	12.2	10.5 (−0.05)	12.9 (0.02)	14.0 (0.05)	18.6 (0.18)
50-54	11.7	9.5 (−0.07)	11.6 (−0.00)	18.2 (0.18)	19.0 (0.20)
55-59	10.5	7.6 (−0.10)	13.3 (0.08)	21.1 (0.29)	21.1 (0.29)
60-64	7.7	4.2 (−0.15)	10.2 (0.09)	20.0 (0.36)	19.0 (0.34)
Relationship status					
Single	44.1	54.2 (0.20)	55.9 (0.24)	28.8 (−0.32)	17.5 (−0.60)
Married	44.0	31.4 (−0.26)	24.6 (−0.42)	51.3 (0.15)	61.1 (0.35)
Divorced	10.8	13.2 (0.07)	18.0 (0.20)	18.2 (0.21)	18.0 (0.21)
Widowed	1.1	1.1 (0.01)	1.6 (0.05)	1.7 (0.06)	3.4 (0.16)
Education					
Basic	28.3	40.2 (0.25)	45.8 (0.37)	32.4 (0.09)	21.9 (−0.15)
Upper secondary	7.9	11.5 (0.12)	6.6 (−0.05)	4.4 (−0.15)	3.5 (−0.19)
Vocational training	35.6	27.9 (−0.17)	32.0 (−0.08)	42.5 (0.14)	40.6 (0.10)
Bachelor's degree	16.0	12.6 (−0.10)	7.8 (−0.26)	13.3 (−0.08)	23.8 (0.20)
Higher education	0.6	0.2 (−0.07)	0.1 (−0.08)	0.2 (−0.07)	0.8 (0.02)
Unknown	11.5	7.7 (−0.13)	7.7 (−0.13)	7.2 (−0.15)	9.3 (−0.07)

Table 2. Characteristics of the Danish Population and Exposed Cohorts, 2012[a]

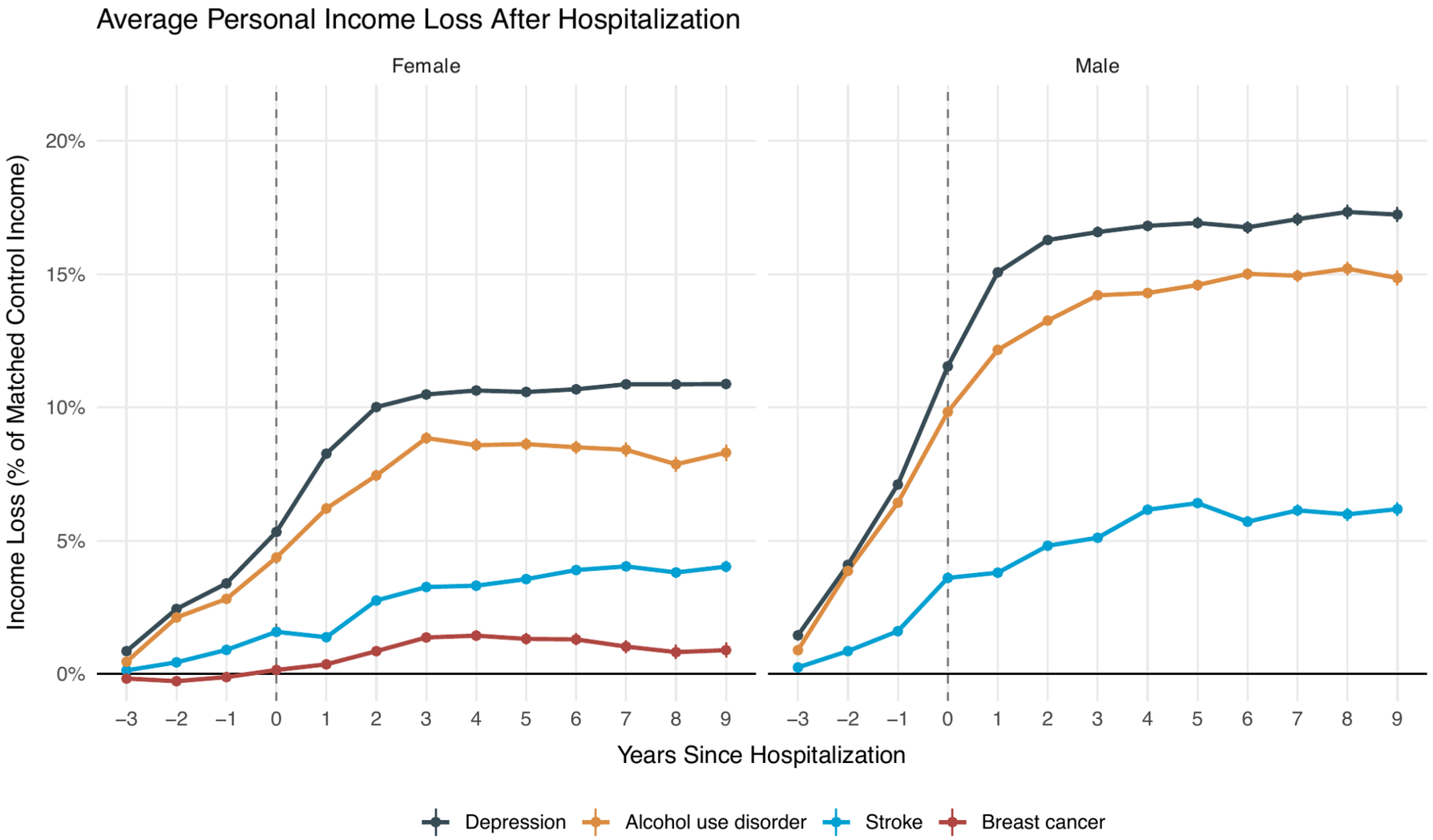
Characteristic	General population (N = 2 911 225), %	Diagnosis, % (SE)			
		Depression (n = 10 284)	Alcohol use disorder (n = 5847)	Stroke (n = 6539)	Breast cancer (n = 2230)
Household size					
1 Person	27.6	35.1 (0.16)	49.4 (0.46)	29.1 (0.03)	19.1% (−0.20)
2 People	29.6	27.4 (−0.05)	24.9 (−0.11)	40.3 (0.23)	43.8% (0.30)
≥3 People	42.8	37.5 (−0.11)	25.7 (−0.37)	30.6 (−0.26)	37.1% (−0.12)
Immigration status					
Danish heritage	86.9	83.3 (−0.10)	89.9 (0.09)	91.5 (0.15)	91.7 (0.15)
First generation	1.7	1.8 (0.01)	1.3 (−0.03)	0.5 (−0.12)	0.5 (−0.11)
Immigrant	11.4	14.8 (0.10)	8.7 (−0.09)	8.0 (−0.11)	7.8 (−0.12)
Baseline income quintile					
1 (Lowest)	18.1	30.2 (0.28)	29.0 (0.26)	9.5 (−0.25)	6.3 (−0.37)
2	13.7	19.9 (0.17)	21.4 (0.20)	18.3 (0.13)	12.1 (−0.05)
3	20.3	23.8 (0.08)	22.8 (0.06)	26.0 (0.14)	25.7 (0.13)
4	24.8	17.3 (−0.18)	16.7 (−0.20)	25.4 (0.01)	28.8 (0.09)
5 (Highest)	23.2	8.9 (−0.40)	10.1 (−0.36)	20.8 (−0.06)	27.1 (0.09)
Labor market attachment					
Employed	69.6	49.0 (−0.43)	46.1 (−0.49)	65.3 (−0.09)	77.2 (0.17)
In school	11.9	19.0 (0.20)	13.3 (0.04)	3.0 (−0.34)	1.1 (−0.45)
On leave from work	1.6	3.6 (0.13)	2.5 (0.07)	2.1 (0.04)	1.4 (−0.01)
Unemployed	1.9	2.8 (0.06)	3.2 (0.09)	2.5 (0.05)	1.7 (−0.01)
Receiving disability benefits	6.7	6.9 (0.01)	14.4 (0.25)	17.2 (0.33)	11.4 (0.17)
Receiving welfare benefits	5.1	13.0 (0.28)	15.6 (0.35)	7.6 (0.10)	4.6 (−0.02)
Other	3.3	5.6 (0.11)	4.9 (0.08)	2.2 (−0.07)	2.5 (−0.05)
Hospital utilization, 5-y average visits					
0	41.8	31.9 (−0.21)	35.6 (−0.13)	27.8 (−0.30)	23.6 (−0.40)
1	19.9	18.2 (−0.04)	17.3 (−0.07)	17.5 (−0.06)	25.6 (0.14)
2	11.8	12.5 (0.02)	12.0 (0.01)	12.4 (0.02)	14.1 (0.07)
3	8.0	9.2 (0.04)	9.1 (0.04)	9.5 (0.05)	10.7 (0.09)
≥4	18.5	28.1 (0.23)	26.0 (0.18)	32.8 (0.33)	26.1 (0.18)

Data visualization

- Questions:
  - What do the different disease populations look like?
  - How do they differ?
  - What are the average outcomes of exposed, control groups?
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Exploration – shiny app

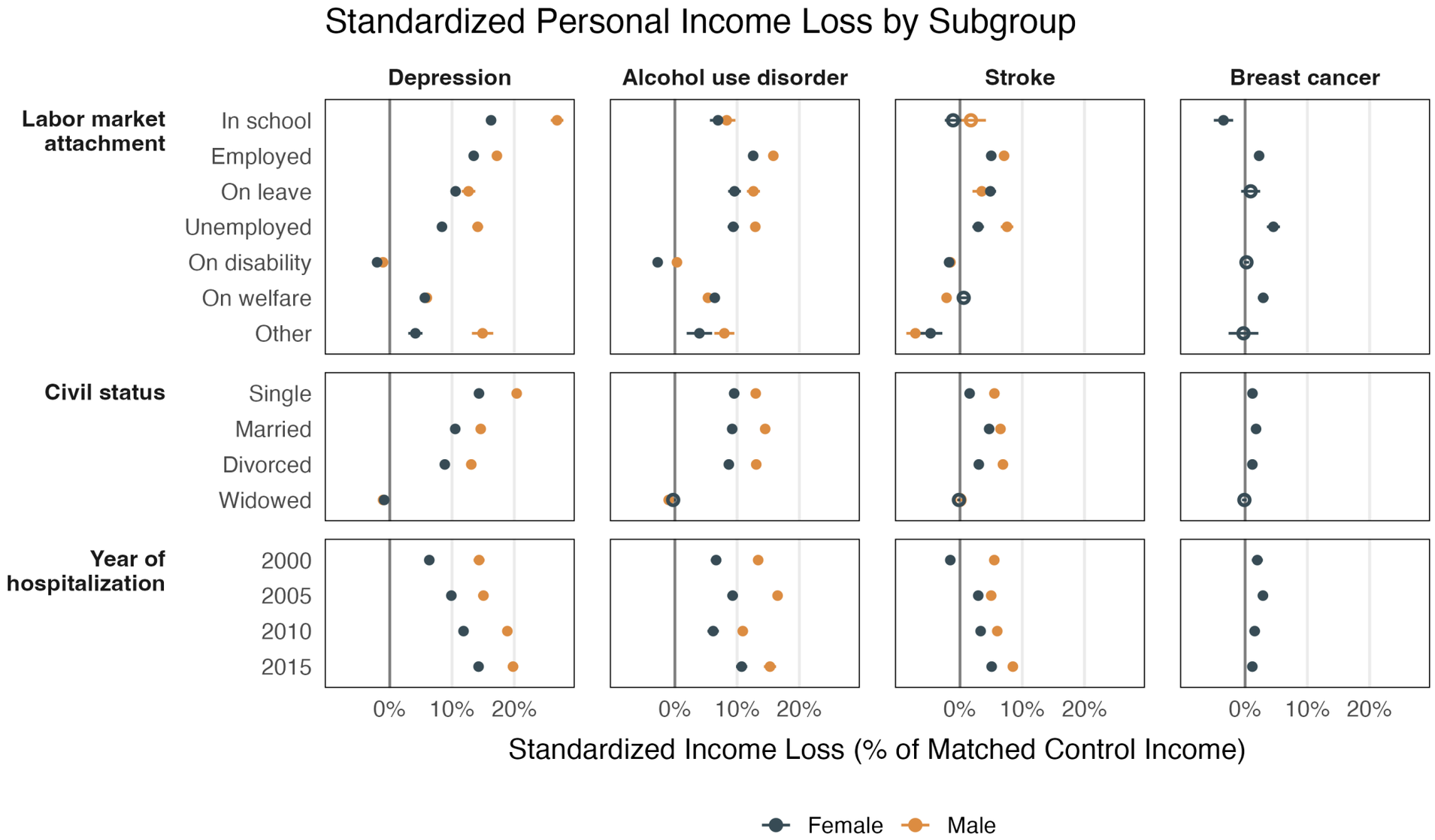
Income losses continue 10 years after diagnosis, not explained by disease risk



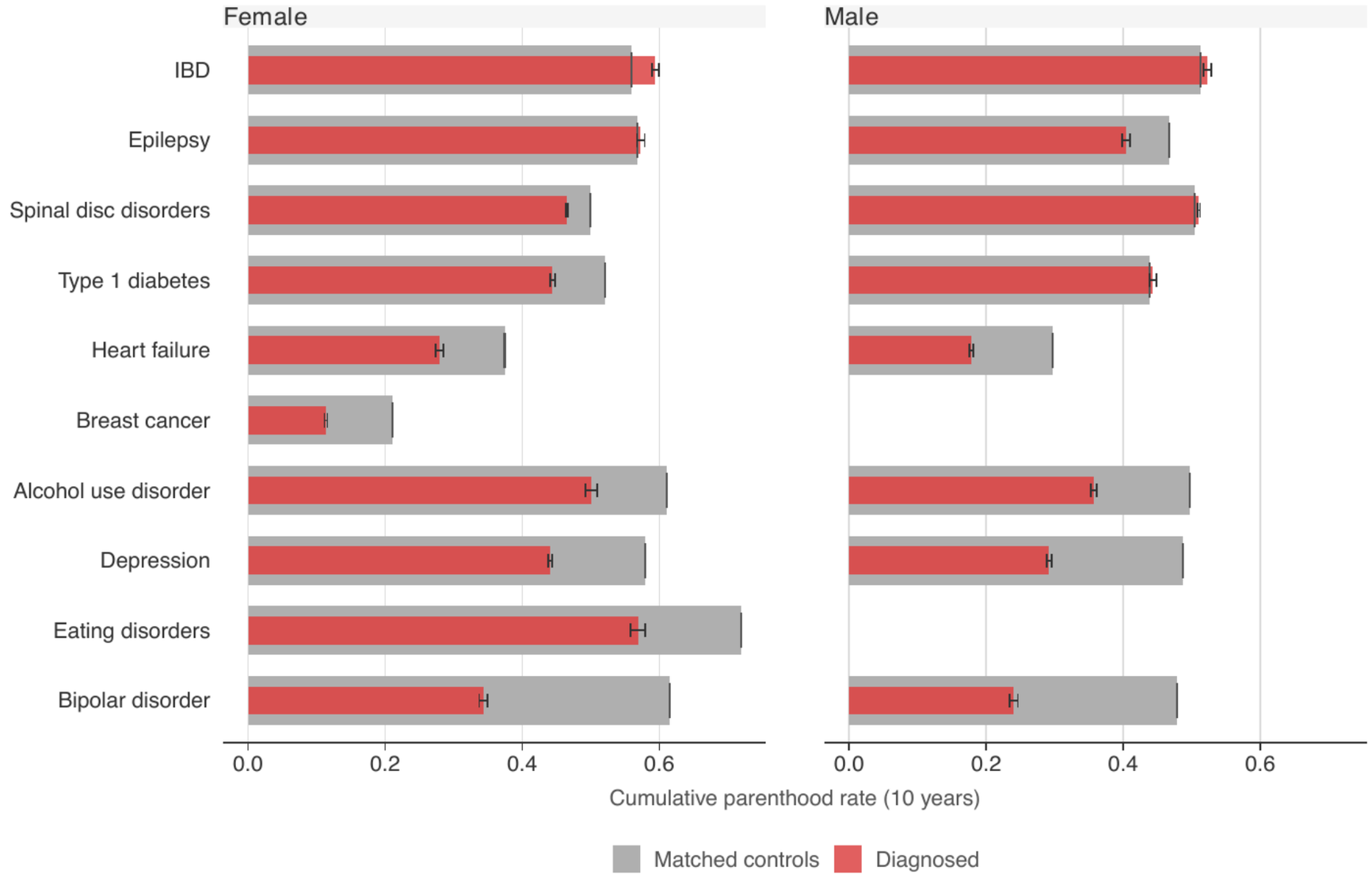
Prime working ages, young males with depression most impacted



Differential loss patterns across disease and demographics

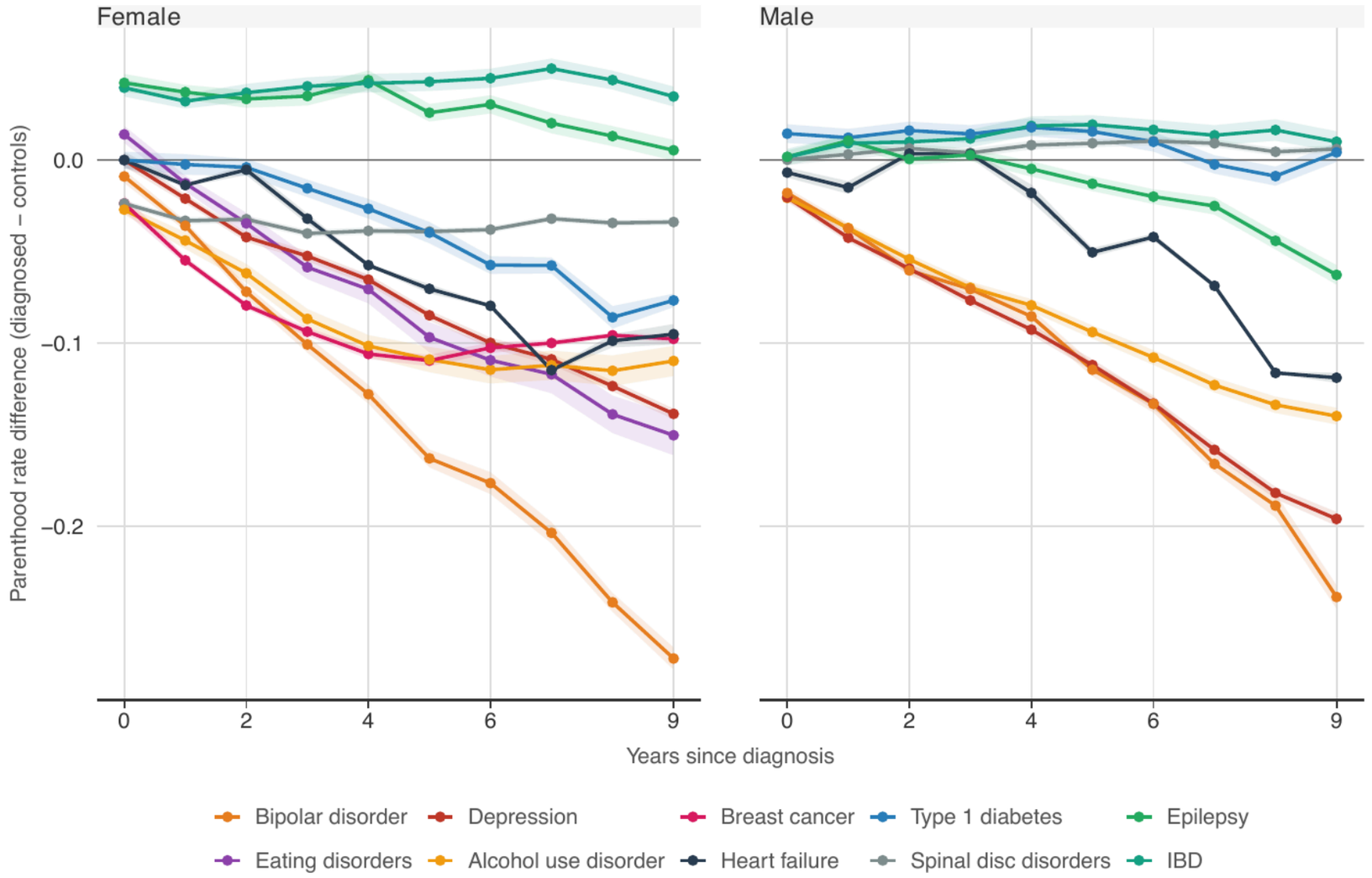


Disease diagnosis and 10y parenthood rates



Widening parenthood gap with time for many diseases

Cumulative 10-year parenthood rates differences over time



Recap

- ExpBoD project – large data management project
  - Iterative
  - Large datasets
- Data interpretation
  - Table 1
  - Exploratory data visualization/analysis
  - Explanatory visualization